

Spatial and spatio-temporal modelling

Structured space-time dependence

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Aug 6, 2026

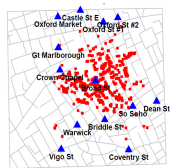
Introduction



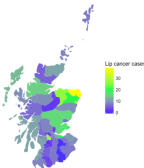
What is spatial modelling?

Disease Mapping

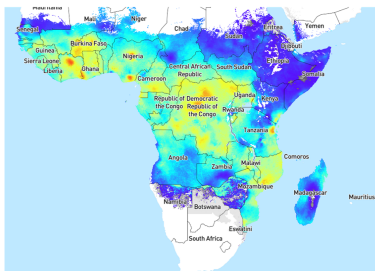
Snow's Cholera London



Scottish lip cancer

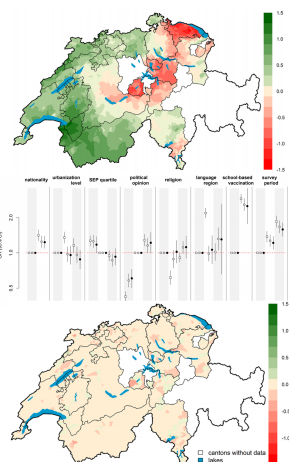


Malaria in Africa in 2007

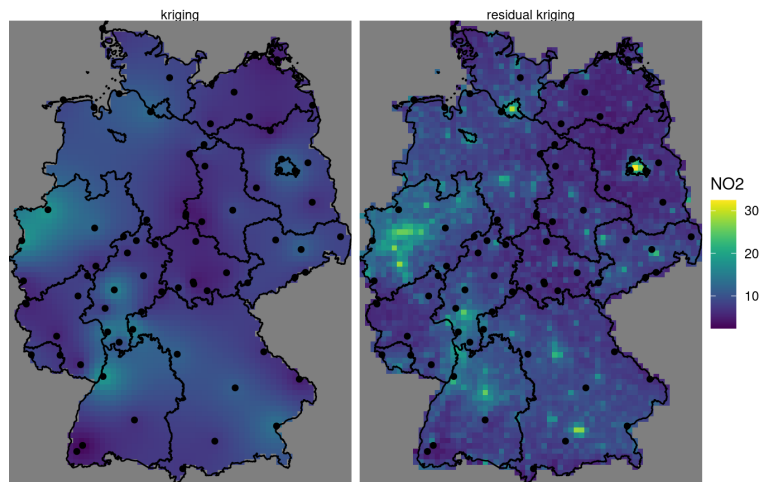


Spatial Regression

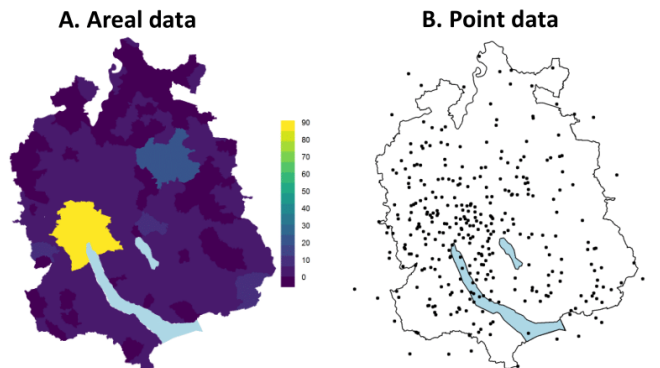
HPV vaccination uptake in Switzerland



It is also geostatistics



Type of data



Why do we need more complex models?

- In disease mapping estimates of disease risk can be unstable, leading to non-meaningful maps.
- Accounting for residual spatial autocorrelation is important to have the correct uncertainty.
- Sometimes they pick up unknown spatial confounding, leading to unbiased estimators.

Structure of the talk

- Disease mapping and smoothing
- Global smoothing
- Local smoothing
- Examples

Disease mapping and smoothing

Disease mapping - background

- To summarise spatial and spatio-temporal variation in disease risk.
- Rare disease and/or small areas: Poisson framework

$$O_i \sim \text{Poisson}(\lambda_i E_i)$$

where E_i is the expected nb of cases and $\lambda_i = \text{SMR}$ in area i .

- Non smoothed estimates of the standardised mortality ratio (SMR):

$$\text{SMR}_i = \frac{O_i}{E_i}, \text{ and } \text{Var}(\text{SMR})_i = \frac{O_i}{E_i^2}$$

Issues with crude RR estimates

Calculating crude (non-smoothed) estimates of the SMR can be problematic:

- SMR is estimated independently: does not account for spatial correlation, i.e. makes no use of SMR estimates in other areas of the map, even though these are likely to be similar.
- They are very imprecise: areas with small E_i have high associated variance.

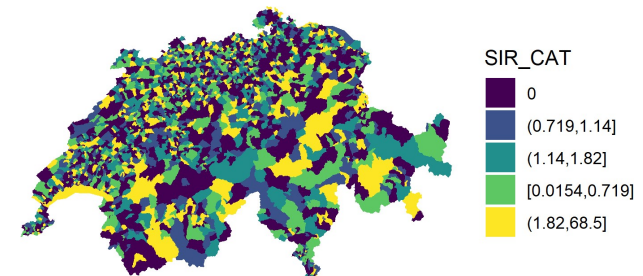
Example with childhood cancers in Switzerland

- Rare cancers (6,000 cases in 31 years)
- Little known about causes, potential clusters



Example with childhood cancers in Switzerland

- Quartiles and 0s a separate category
- Is this map informative?

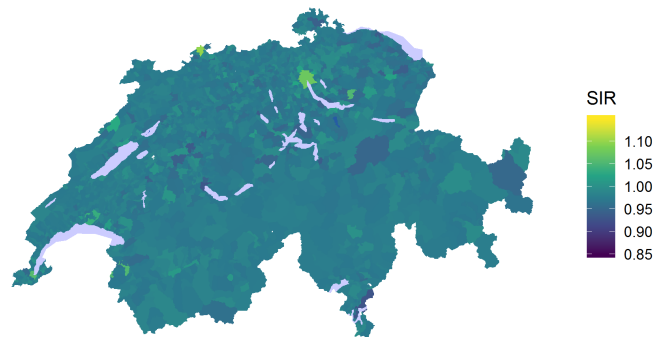


Hierarchical modelling to smooth rates

- **Globally:** Smooth the rates towards the global mean.
- **Locally:** Smooth the rates towards the local mean, using information about the adjacency structure.
- **Globally + Locally:** Smooth the rates using information about the global mean and the adjacency structure.

Example with childhood cancers in Switzerland

- Plot $\exp(\alpha + \theta_i)$



Global smoothing

$$\begin{aligned} O_i &\sim \text{Poisson}(\lambda_i E_i) \\ \log \lambda_i &= \alpha + \theta_i \\ \theta_i &\sim \text{Normal}(0, \sigma_\theta^2) \end{aligned}$$

θ_i : area specific random effects to account for overdispersion

- Excess variation in the observed counts due to random noise.
- Latent variable which captures the effects of unknown or unmeasured area level covariates.

$\exp(\theta_i)$: RR in area i, relative to the overall SMR/SIR

Issues with global smoothing

- Poisson-logNormal model assumes identically distributed and independent data.
- Data that occur close together in space are likely to be correlated (Dependence assumption more realistic).
- Ignoring spatial dependence can lead to biased and inefficient inference.
- Interest in estimating the relationship between location and outcome.

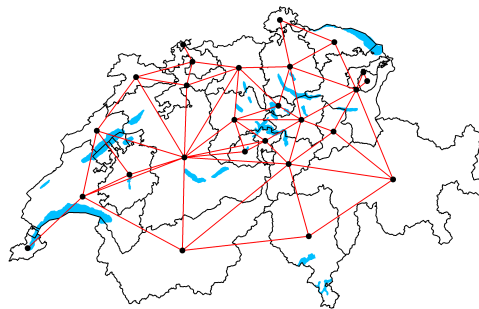
Issues with global smoothing

Local smoothing: prior distribution for the random effects allowing for spatial correlation to help uncover spatial patterns.

Example: Queen contiguity in Switzerland (Cantons)

Let ∂_i = set of areas adjacent to i

$$w = \begin{cases} 1, & \text{if } j \in \partial_i \\ 0 & \text{otherwise} \end{cases}$$



Specify a distribution to account for information

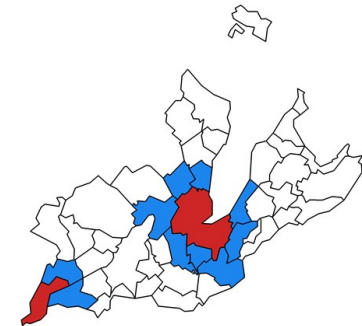
- Specify a distribution which takes into account information on neighbouring areas (**adjacency matrix**).
- Rule for determining the neighbours of each area: most common based on common boundary.
- Estimate **spatial random effect** for each area as if we knew the values of the spatial random effects in neighbouring areas.
- Use of conditional autoregressive distributions. (we are conditioning on knowing the neighbours)

Intrinsic ICAR model (Besag et al. 1991)

$$\phi \sim \text{ICAR}(W, \sigma_\phi^2) \Leftrightarrow \phi_i | \phi_{j \neq i} \sim \text{Normal}(\phi_i, \sigma_\phi^2)$$

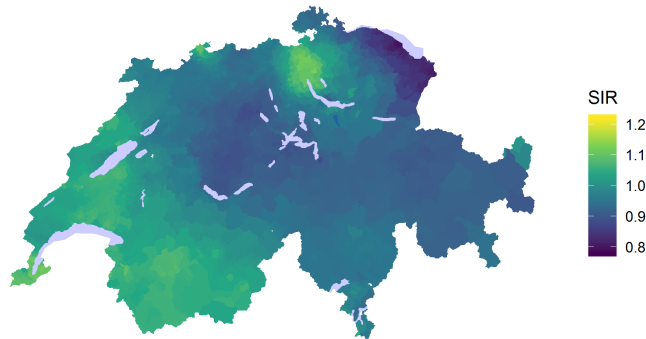
$$E(\phi_i | \phi_j) = \sum_{j \in \partial_i} \phi_j / n_i$$

$$\text{Var}(\phi_i | \phi_j) = \sigma_\phi^2 / n_i$$



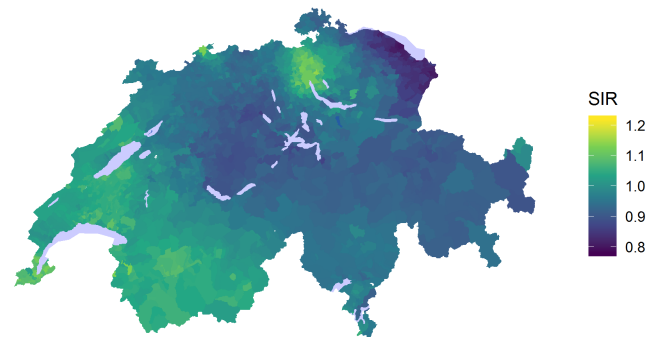
Example with childhood cancers in Switzerland (cont.)

- Plot $\exp(\alpha + \phi_i)$



Example with childhood cancers in Switzerland (cont)

- Plot $\exp(\alpha + \theta_i + \phi_i)$



The Besag-York-Mollie model

Local+Global smoothing:

$$\begin{aligned} O_i &\sim \text{Poisson}(\lambda_i E_i) \\ \log \lambda_i &= \alpha + \theta_i + \phi_i \\ \theta_i &\sim \text{Normal}(0, \sigma_\theta^2) \\ \phi &\sim \text{ICAR}(W, \sigma_\phi^2) \end{aligned}$$

priors for α , σ_θ^2 and σ_ϕ^2

Use BYM with care

- ICAR model defined above is improper: the overall mean of the ϕ_i is not defined: **sum-to-zero constraints**.
- Inference may be sensitive to choice of hyperprior for the random effects variance or precision parameters.
- Other known issues we have not discussed about: different interpretation of the variances (marginal conditional), scaling the covariance matrix, etc.

Other CAR models

Let $\log \lambda_i = \alpha + b_i$:

- The proper CAR

$$b_i | b_j \sim \text{Normal}(\rho \sum_{j \in d_i} b_j / n_i, \sigma_b^2 / n_i)$$

- Leroux CAR prior

$$b_i | b_j \sim \text{Normal}\left(\frac{\rho \sum_{j \in d_i} b_j}{\rho n_i + (1 - \rho)}, \frac{\sigma_b^2}{\rho n_i + (1 - \rho)}\right)$$

Posterior Probability

- Mapping the posterior mean RR (or residual RR) does not make full use of the output of the Bayesian analysis that provides, for each area, samples from the whole posterior distribution of the relative risk.
- Mapping the probability that a RR is greater than a specified threshold of interest has been proposed by several authors (e.g. Clayton and Bernardinelli (1992)).
- Very effective method to identify areas characterised by elevated risk.

Other CAR models

- BYM2

$$\begin{aligned} b &= \sigma_b * (\sqrt{1 - \rho} \theta + \sqrt{\rho} \phi) \\ \theta &\sim \text{Normal}(\sigma_\theta^2) \\ \phi &\sim \text{ICAR}(W^*, \sigma_\phi^2) \end{aligned}$$

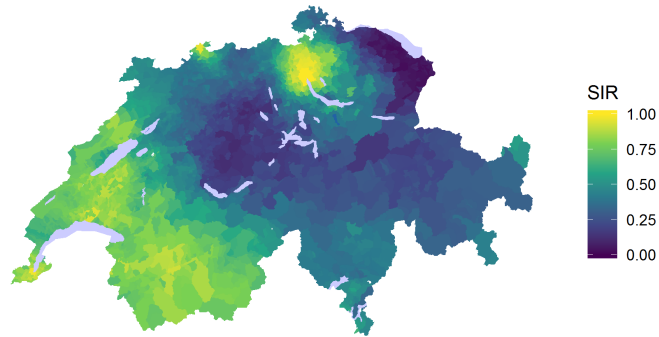
- Adaptive CARs

How to classify areas as having elevated risk

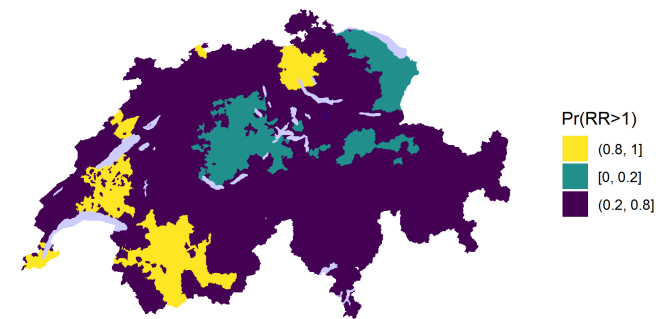
- We define the decision rule $D(c, RR_0)$, which depends:
 - on a cutoff probability c
 - a reference threshold RR_0
- Area i is classified as having an elevated risk according to

$$D(c, RR_0) : \text{Prob}(RR_i > RR_0) > c$$
- Recommended values $c = 0.8$ and $RR_0 = 1$ (Richardson et al. 2004)
- Posterior probability of interest: $\text{Prob}(\exp(\theta_i + \phi_i) > 1)$

Example with childhood cancers in Switzerland



Example with childhood cancers in Switzerland



Ecological regression

Extending to ecological regression

Straightforward extension of the BYM (read any CAR) model:

$$\begin{aligned} O_i &\sim \text{Poisson}(\lambda_i E_i) \\ \log \lambda_i &= \alpha + \beta X_i + \theta_i + \phi_i \\ \theta_i &\sim \text{Normal}(0, \sigma_\theta^2) \\ \phi &\sim \text{ICAR}(W, \sigma_\phi^2) \end{aligned}$$

- X area-level covariate of interest
- β : parameter associated with the covariate (assigned a prior)

Interpretation of the parameters

- $\exp(\beta)$ is the relative risk of disease associated with a unit increase in exposure x (when X continuous).
- $\exp(\theta_i + \phi_i)$ is the residual or adjusted relative risk of disease in area i after accounting for the effect of x .

Extension to several variables

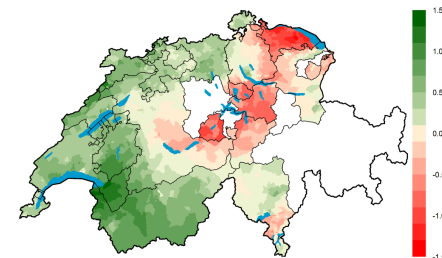
- is the relative risk of disease associated with a unit increase in exposure x , after adjustment for z
- Same for z , after adjustment for x
- is the adjusted relative risk of disease in area i after accounting for the effects of measured covariates

Example with HPV vaccination in Switzerland

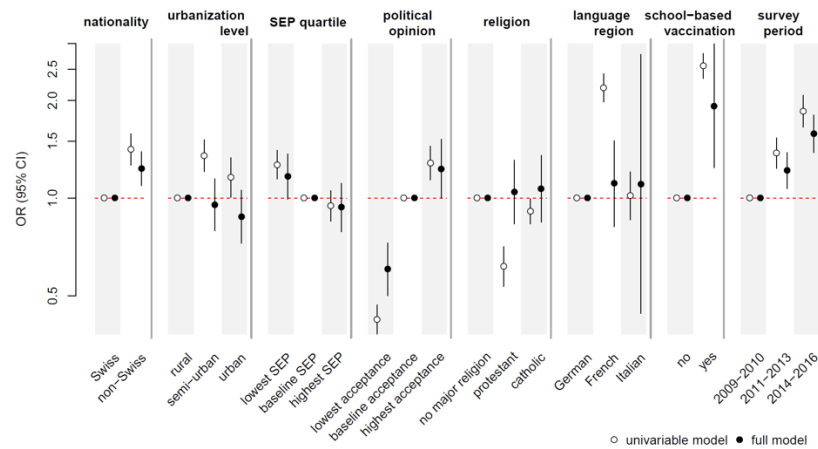
- Covariates related with age, sex, period of survey, deprivation, religion, language region, etc.

HPV vaccination uptake

- Model without any covariates just the BYM prior
- Plot (on the log-odds)

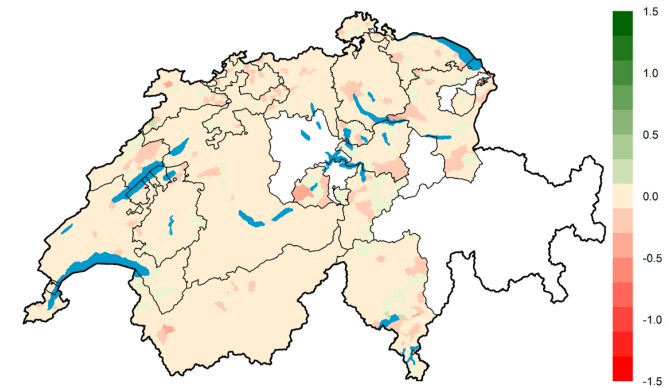


Effect of covariates



Building complex spatiotemporal models

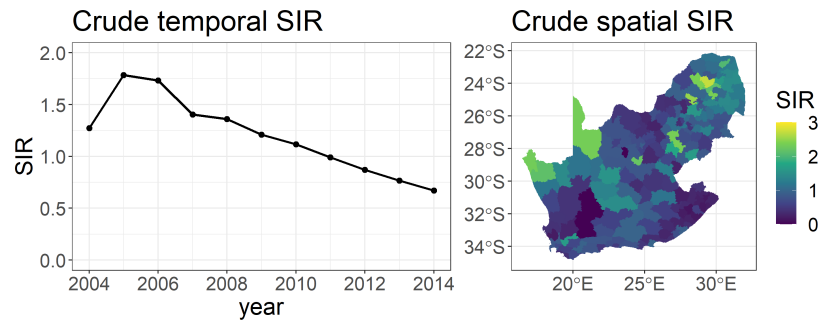
Residual spatial autocorrelation



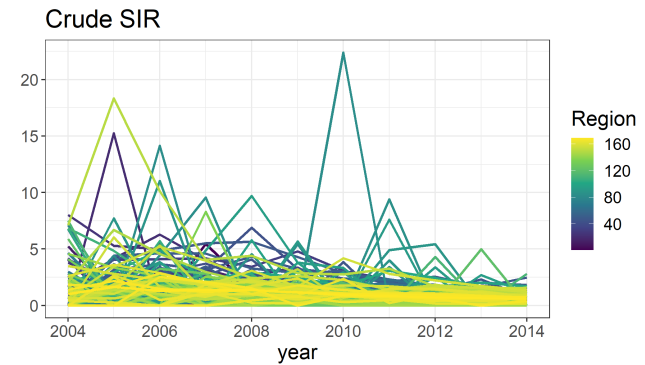
Cervical cancers among HIV positive women

- Global cohort composed of 1,700,000 persons living with HIV/AIDS.
- SAM study, evaluating HIV related cancers in South Africa.
- 17% of adult women in South Africa over the age of 25 years are living with HIV.
- High burden of cervical cancer: 43.1 cases per 100,000.
- 5 times more likely to develop cervical cancer.

Exploratory analysis of the crude rates



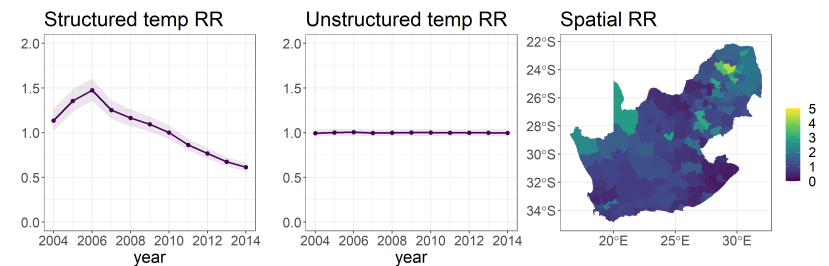
Exploratory analysis of the crude rates (cont.)



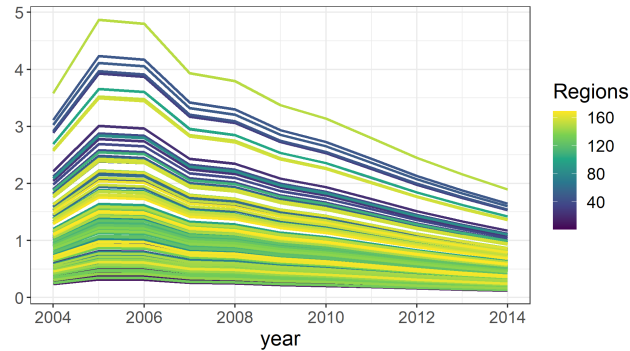
A spatiotemporal model

Similar with the BYM model in space is the temporal unstructured and the temporal structured random effect, capturing the temporal autocorrelation.

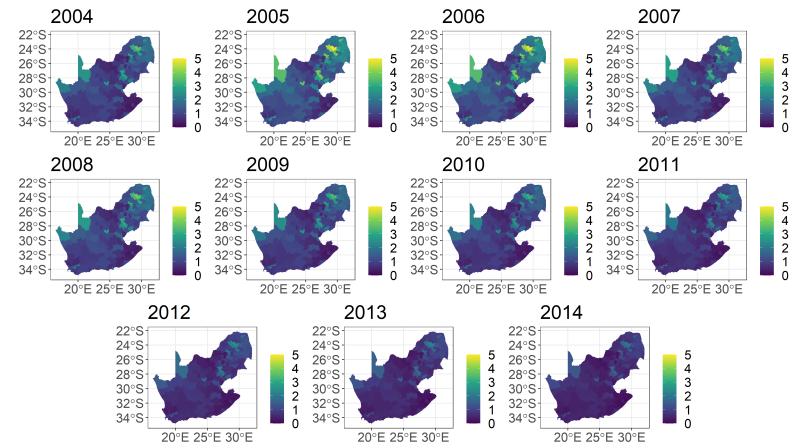
Space and time separately



Space-time (Spaghetti plot)



Space-time (Space by year)



Space-time interactions

Interaction	Parameters
type 1	
type 2	
type 3	
type 4	

Summary

- Introduction to disease mapping and CAR priors
- Extension to ecological regression and relevant interpretations
- Separable spatiotemporal models
- Higher spatiotemporal interactions
- Examples using childhood cancers, HPV vaccination, and cervical cancers in South Africa

Next : Coding these models in NIMBLE.

Getting ready for the lab

The lab for this session

- This goal of this lab is to use [NIMBLE](#) to carry out a disease mapping study.
- During this lab session, we will:
 1. Explore ways of visualizing spatial data;
 2. Define the neighborhood matrix in [R](#);
 3. Fit and interpret the BYM model in [NIMBLE](#); and
 4. Perform spatial ecological regression

Questions?